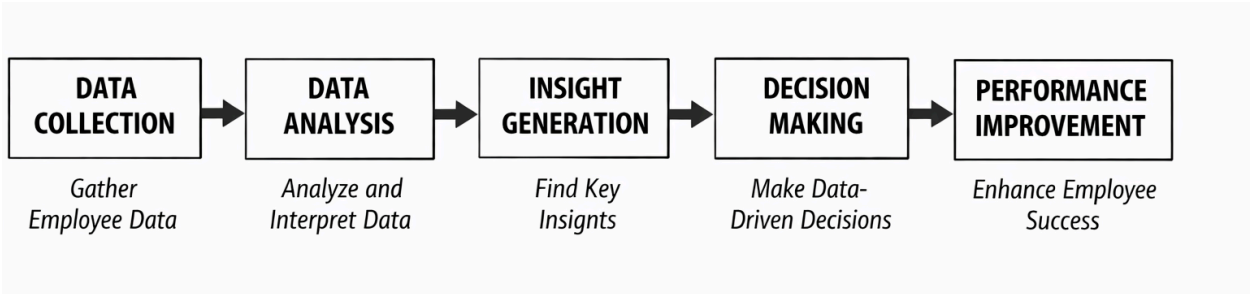
HUMAN RESOURCE ANALYTICS TO IMPROVE EMPLOYEE PERFORMANCE

Human Resource (HR) Analytics is the process of **collecting, analyzing, and using employee data** to improve workforce performance, productivity, and decision-making.

HR ANALYTICS PROCESS



HOW HR ANALYTICS IMPROVES EMPLOYEE PERFORMANCE

1. PERFORMANCE MEASUREMENT

Collect employee performance data such as:

- Productivity
- Task completion rate
- Quality of work

Analytics Role:

- Identify high and low performers
- Compare performance across teams
- Set measurable performance standards

---

2. IDENTIFYING SKILL GAPS

Find missing skills required for job roles

Analytics Role:

- Analyze training records and performance data
- Identify weak areas in employees
- Suggest required skill improvements

---

3. TRAINING AND DEVELOPMENT

Provide proper training programs

Analytics Role:

- Recommend personalized training
- Measure training effectiveness
- Improve employee knowledge and efficiency

---

4. EMPLOYEE ENGAGEMENT ANALYSIS

Understand employee satisfaction and motivation

Analytics Role:

- Analyze survey data and feedback
- Identify disengaged employees
- Improve workplace environment

---

5. PREDICTIVE ANALYTICS FOR PERFORMANCE

Predict future performance using data

Analytics Role:

- Identify employees at risk of poor performance
- Predict future productivity trends
- Take early corrective actions

---

6. PERFORMANCE APPRAISAL IMPROVEMENT

Improve evaluation system

Analytics Role:

- Use data instead of manual judgment
- Ensure fair and unbiased appraisal
- Provide accurate performance ratings

---

7. REDUCING EMPLOYEE TURNOVER

Prevent employees from leaving

Analytics Role:

- Identify reasons for attrition
- Predict employees likely to leave
- Take retention actions

ANALYTICS TECHNIQUES USED IN HR

- **Descriptive Analytics** → What happened
  - Analyzes past employee data to understand what has already happened in the organization.
  - It uses reports, dashboards, and summaries to show trends like attendance, performance, and turnover.

- **Diagnostic Analytics** → Why it happened
  - Examines data to find the reasons behind employee performance or issues.
  - It helps identify causes like low motivation, lack of skills, or poor work environment.

- **Predictive Analytics** → What will happen
  - Uses historical data and models to predict future employee behavior and performance.
  - It helps forecast outcomes like employee attrition, productivity, and training needs.

- **Prescriptive Analytics** → What action to take
  - Suggests the best actions to improve employee performance based on data insights.
  - It helps in decision-making like training programs, incentives, and workforce planning.

BENEFITS OF HR ANALYTICS

- Improves employee performance
- Increases productivity
- Better decision making
- Reduces employee turnover
- Enhances employee satisfaction
- Supports strategic planning

PREDICTIVE ANALYTICS–BASED MARKETING & SALES CAMPAIGN

A predictive analytics–based marketing and sales campaign uses **historical data, statistical models, and machine learning** to **anticipate customer behaviour**, target the right audience, and maximize business outcomes like sales, profit, and customer satisfaction.

---

1. DATA COLLECTION AND INTEGRATION

- Collect large amount of customer data from multiple sources:
  - Transaction data (purchase history)
  - Website and app usage
  - Social media interactions
  - Customer feedback and CRM data

Application of Analytics:

- Integrate structured and unstructured data
- Clean and organize data for analysis
- Build a complete 360° customer profile

---

2. CUSTOMER SEGMENTATION

- Divide customers into meaningful groups

Examples:

- High-value customers
- Price-sensitive customers
- Frequent buyers
- New or inactive customers

Application of Predictive Analytics:

- Use clustering algorithms to group customers
- Identify profitable and high-potential segments
- Focus marketing efforts on important segments

---

3. CUSTOMER BEHAVIOUR PREDICTION

- Predict future actions of customers

Predictions include:

- Probability of purchase
- Product preferences
- Customer lifetime value (CLV)
- Churn probability

Application of Predictive Analytics:

- Use machine learning models like regression, classification
- Forecast buying patterns and trends
- Identify customers who are likely to respond

---

4. PERSONALIZED MARKETING STRATEGY

- Design customized marketing campaigns

Examples:

- Personalized emails and SMS
- Product recommendations
- Special discounts and offers

Application of Predictive Analytics:

- Recommend products based on past behaviour
- Send offers at the right time
- Improve customer engagement and response

---

5. SALES OPTIMIZATION

- Improve efficiency of sales process

Application of Predictive Analytics:

- Lead scoring to identify high-potential customers
- Predict best pricing and discount strategies
- Cross-selling and up-selling recommendations
- Improve sales conversion rate

---

6. CAMPAIGN EXECUTION

- Implement campaigns through multiple channels:
    - Email marketing
    - Social media marketing
    - Online advertisements
    - Direct sales

Application of Analytics:

- Monitor campaign in real-time
- Track customer response and engagement
- Adjust campaign strategies quickly

---

7. PERFORMANCE MEASUREMENT

- Evaluate success of the campaign

Key Metrics:

- Conversion rate
- Click-through rate (CTR)
- Customer acquisition cost (CAC)
- Return on Investment (ROI)

Application of Predictive Analytics:

- Identify which campaigns perform better
- Analyze customer response patterns
- Optimize marketing and sales strategies

8. CONTINUOUS IMPROVEMENT AND FEEDBACK

- Use feedback to improve future campaigns

Application of Predictive Analytics:

- Update models with new data
- Improve prediction accuracy
- Make campaigns more effective over time

ANALYTICS TECHNIQUES USED

- Predictive modeling (regression, classification)
- Machine learning algorithms
- Customer segmentation (clustering)
- Recommendation systems
- Lead scoring models

BENEFITS OF USING PREDICTIVE ANALYTICS

- Better understanding of customer behaviour
- Increased sales and revenue
- Improved customer targeting and personalization
- Reduced marketing cost
- Higher customer satisfaction and retention
- Better strategic decision-making

SIMPLE LINEAR REGRESSION FOR SALES PREDICTION (ML PREDICTIVE ANALYSIS)

Simple Linear Regression is a **statistical and machine learning technique** used to model the relationship between one independent variable (time) and one dependent variable (sales) to **predict future sales values**.

$$y = a + bx$$

Where:

- **y** → Predicted sales
- **x** → Time (month)
- **a** → Intercept (sales when x = 0)
- **b** → Slope (change in sales per month)

HISTORICAL DATA (GIVEN)

Month (x)	Sales (y)
1	100
2	120
3	130
4	150
5	170

STEP 1: CALCULATE REQUIRED VALUES

x	y	x^2	xy
1	100	1	100
2	120	4	240
3	130	9	390
4	150	16	600
5	170	25	850

- $\Sigma x = 15$
- $\Sigma y = 670$
- $\Sigma x^2 = 55$
- $\Sigma xy = 2180$
- $n = 5$

STEP 2: CALCULATE SLOPE (b)

$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$

Substitute values:

$$b = \frac{5(2180) - (15)(670)}{5(55) - (15)^2}$$

$$b = \frac{10900 - 10050}{275 - 225} = \frac{850}{50} = 17$$

STEP 3: CALCULATE INTERCEPT (a)

$a = \frac{\sum y - b\sum x}{n}$

$$a = \frac{670 - (17)(15)}{5} = \frac{670 - 255}{5} = \frac{415}{5} = 83$$

STEP 4: FORM REGRESSION EQUATION

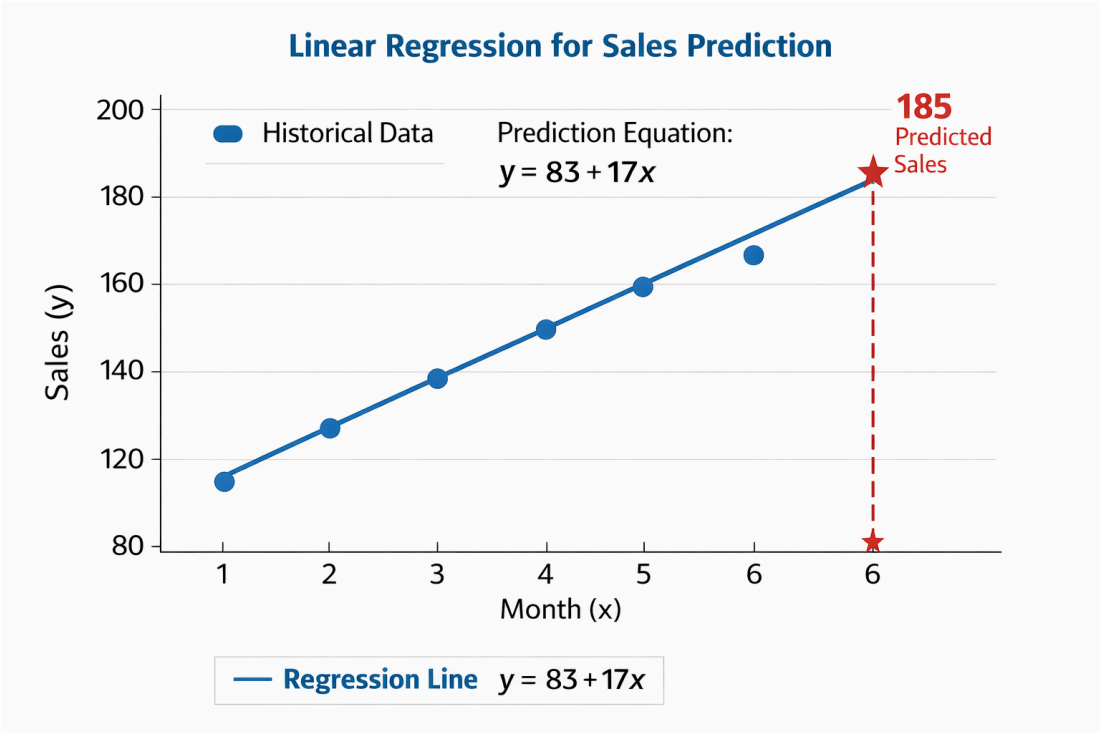
$$y = 83 + 17x$$

STEP 5: PREDICT NEXT MONTH SALES

For next month (x = 6):

$$y = 83 + 17(6) = 83 + 102 = 185$$

Predicted Sales = 185 units



ROLE IN MACHINE LEARNING PREDICTIVE ANALYSIS

- Learns pattern from historical data
- Predicts future sales values
- Helps in demand forecasting and planning
- Supports data-driven business decisions

ADVANTAGES

- Simple and easy to calculate
- Requires less data
- Easy to interpret results
- Useful for short-term prediction